

JOURNAL CLUB

NATURE Trial

Mechanical thrombectomy with enVast in STEMI with large thrombus burden — a randomised superiority trial (ESC Congress 2026)

Presented by

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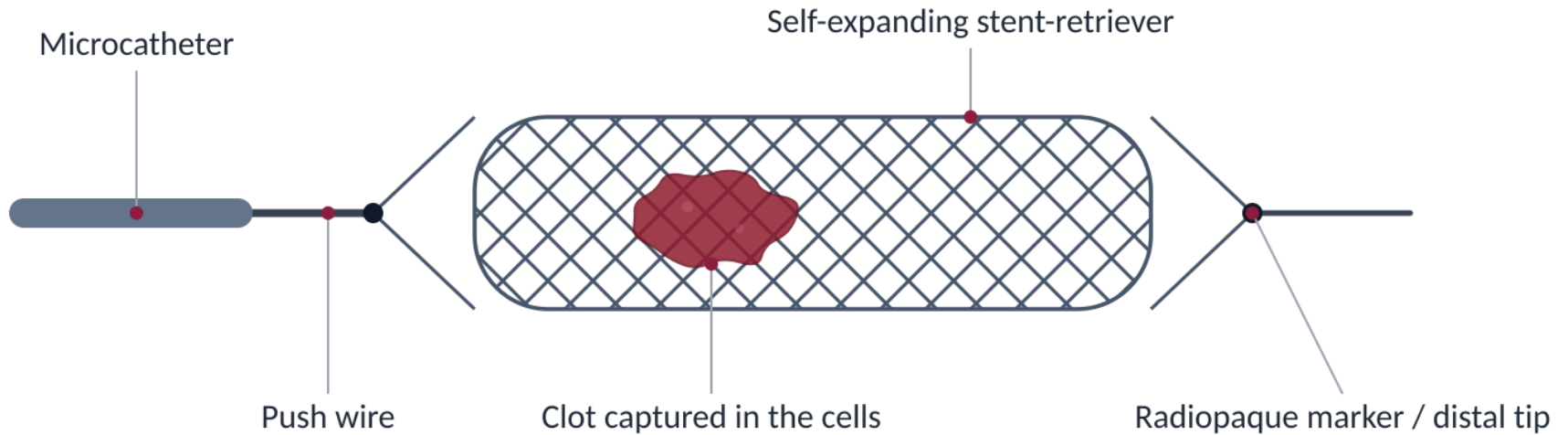
Background: why a large thrombus matters



What we already know

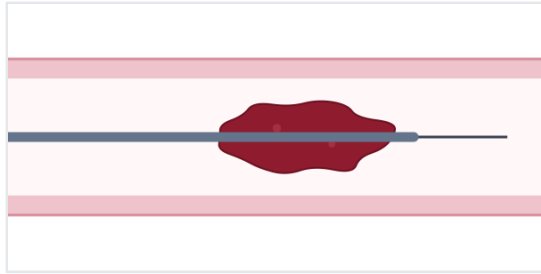
- TASTE (n = 7,244): routine aspiration did not reduce mortality
- TOTAL (n = 10,732): no clinical benefit, and more strokes at 30 days
- ESC guidelines: routine thrombus aspiration is not recommended (Class III)
- Question: can a stent-retriever remove clot better and more safely?

The enVast device



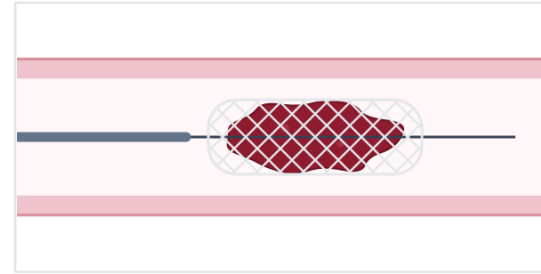
Schematic illustration (not to scale). Coronary stent-retriever by Vesalio, derived from the NeVa neurovascular thrombectomy platform.

The procedure step by step



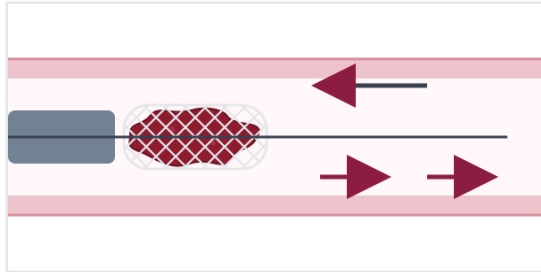
1 Cross

Wire and microcatheter pass through the clot



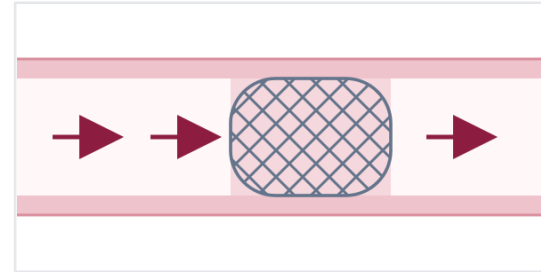
2 Deploy

Retriever opens inside the clot and holds it



3 Retrieve

Device and clot pulled back into the guide

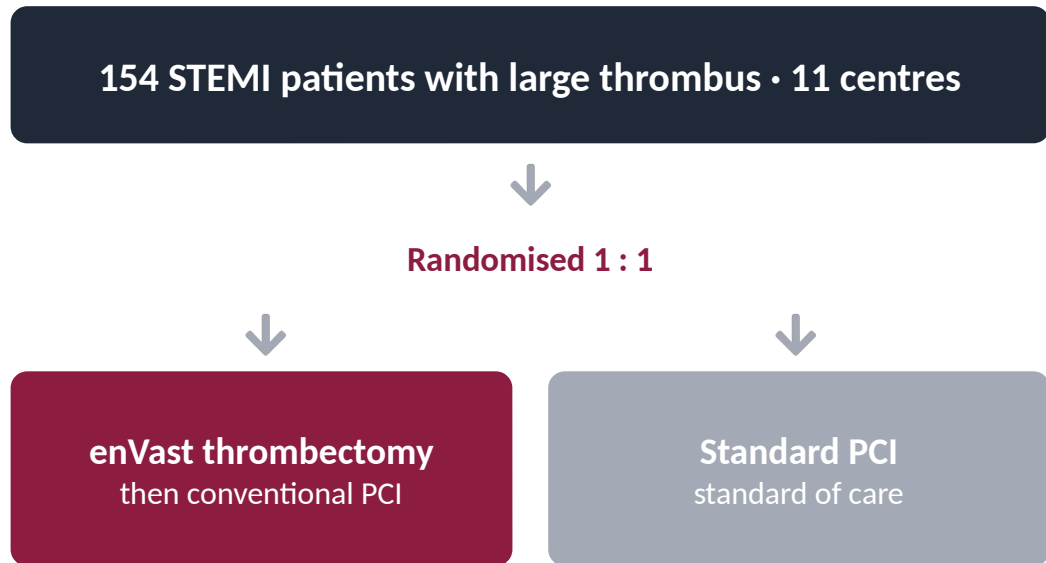


4 Stent

Conventional PCI; flow restored

Schematic illustrations (not to scale).

Trial design



Type	Prospective, multicentre, randomised
Hypothesis	Superiority on infarct size
PI	Prof. Marco Valgimigli, Lugano
Sponsor	Vesalio
Registry	NCT04969471

Inclusion criteria

- ✓ Age ≥ 18 years
- ✓ Chest pain > 20 min with ST elevation ≥ 1 mm in ≥ 2 contiguous leads, or infero-lateral MI with ST depression ≥ 1 mm in ≥ 2 of V1–V3 and a positive terminal T wave
- ✓ TIMI thrombus grade ≥ 3 in the infarct-related artery (re-confirmed after wiring if TIMI 0 flow)
- ✓ Intervention started within 8 hours of symptom onset
- ✓ Informed consent before the procedure

Key point

TIMI thrombus grade ≥ 3

A definite clot longer than half the vessel diameter. These patients are the most likely to embolise clot during PCI, so they have the most to gain from thrombectomy.

Exclusion criteria

- ✕ Unconscious patient
- ✕ Infarct-related artery < 2.5 mm (visual estimate)
- ✕ Severe calcification or extreme tortuosity at or proximal to the culprit lesion
- ✕ Stent thrombosis as the culprit lesion
- ✕ Previous MI in the same territory
- ✕ Women of child-bearing potential
- ✕ Participation in another interventional trial

Why these exclusions?

- Small, calcified or tortuous vessels: hard to deliver a retriever safely
- Stent thrombosis and prior MI in the same territory would confound infarct size

Endpoints



Primary

Infarct size by CK-MB area under the curve



Secondary

Infarct size on cardiac MRI (% of LV mass) at day 3



Safety

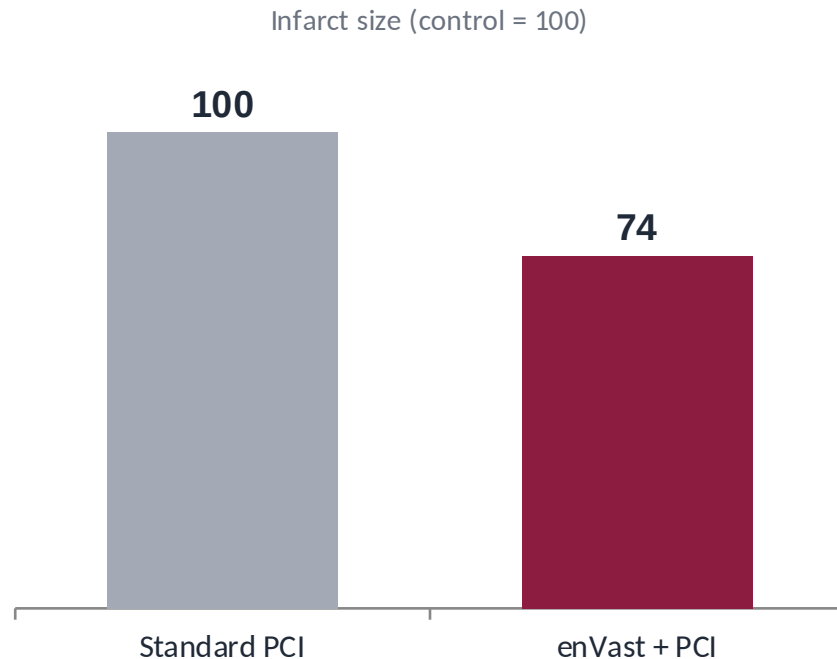
Death, stroke and MACE at 30 days

Primary endpoint: met

26%

relative reduction in infarct size (CK-MB AUC)
with enVast versus standard PCI

P = 0.001



Secondary endpoint: cardiac MRI

CK-MB AUC (primary)

26% smaller

Standard PCI



enVast + PCI

Significant, $P = 0.001$

Cardiac MRI at day 3 (secondary)

25% smaller

Standard PCI

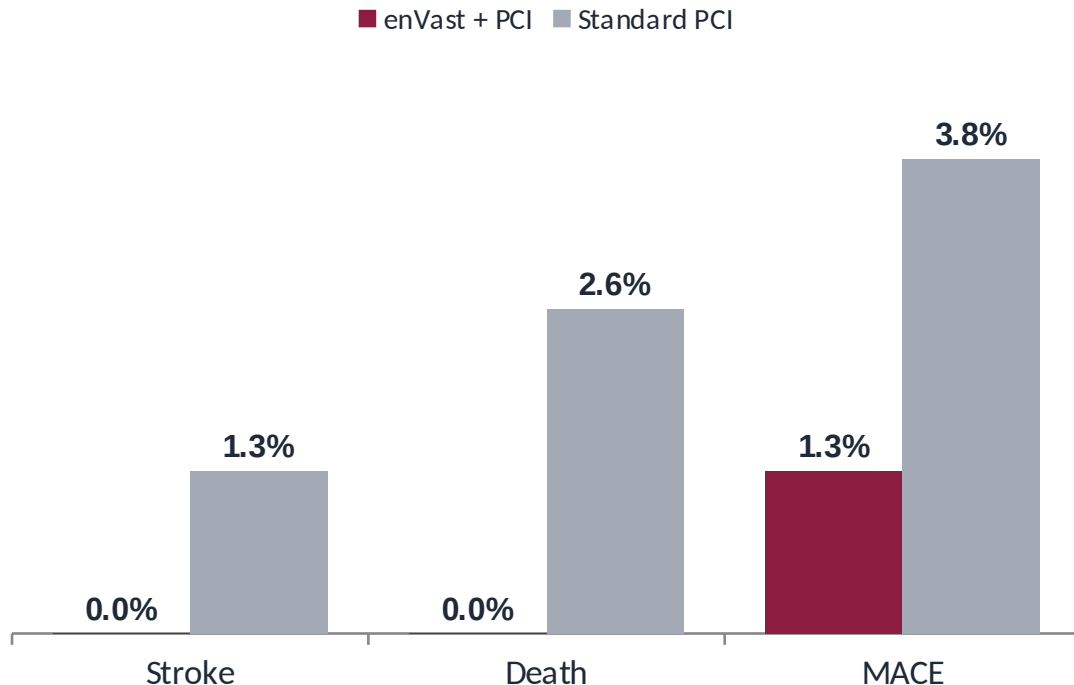


enVast + PCI

Trend in the same direction

Two different measures, a blood test and imaging, point the same way.

Safety at 30 days



No strokes and no deaths with enVast

- Important because aspiration in TOTAL increased stroke
- Very few events overall
- The trial was not powered for clinical outcomes

Strengths



Randomised, multicentre

11 centres, 1:1 allocation



Right population

Only patients with a large thrombus



Consistent results

CK-MB and cardiac MRI agree



Safe early course

No stroke or death with enVast at 30 days

Limitations

1

Small trial

Only 154 patients

2

Surrogate endpoint

Infarct size, not death or heart failure

3

Operators not blinded

Device use cannot be hidden

4

MRI only a trend

Industry-funded; needs independent confirmation

Conclusion

In STEMI with a large thrombus, enVast thrombectomy before PCI was superior to standard PCI in reducing infarct size, with no early safety concern.

Today

Promising for selected large-thrombus cases, but not yet routine. Guidelines are unchanged.

Next

A larger trial with hard outcomes: death, heart failure and stroke.

Take-home for our cath lab

01

Grade the thrombus (TIMI thrombus grade) in every STEMI

02

Routine manual aspiration is still not recommended

03

Stent-retriever thrombectomy is promising for large thrombus, but awaits outcome data

References

1. Valgimigli M. NATURE trial. Late-Breaking Clinical Trial, ESC Congress 2026, Munich.
2. Landi A, et al. Mechanical thrombectomy in STEMI with large thrombus burden: design and rationale of the NATURE trial. Cardiovasc Revasc Med. 2026.
3. ClinicalTrials.gov NCT04969471: NATURE (enVast as an adjunct to PPCI in STEMI).
4. Fröbert O, et al. Thrombus aspiration during STEMI (TASTE). N Engl J Med. 2013.
5. Jolly SS, et al. Primary PCI with or without routine manual thrombectomy (TOTAL). N Engl J Med. 2015.
6. Byrne RA, et al. 2023 ESC Guidelines for the management of acute coronary syndromes. Eur Heart J. 2023.

Thank you

Questions and discussion